

M-Sequences from Galois Fields*

Just enough finite field theory to understand some properties of maximum length sequences (hereafter called M-sequences). This document is not concerned with proofs or rigour; it is very much a ‘roughly speaking’ take on the subject. Knowledge of basic finite group theory is assumed and the presentation leans heavily on David Forney’s MIT course *Principles of Digital Communication II*.

I. FINITE FIELD THEORY

A *field* $\mathbb{F}$ is set S endowed with operations $+$ and $\cdot$ such that $\{S, +\}$ and $\{S \setminus \{0\}, \cdot\}$ are abelian groups with identity elements 0 and 1 respectively. There is also the constraint that the operations mesh nicely with each other $a \cdot (b + c) = a \cdot b + a \cdot c$. Examples of infinite fields are $\mathbb{R}, \mathbb{C}$ and $\mathbb{Q}$. We shall be concerned with finite fields, the simplest of which are the numbers mod- p for p prime. That is $\{0, 1, \dots, p-1\}$ under normal addition and multiplication modulo p . We denote this $\mathbb{F}_p$. In fact, as we shall discuss later, all finite fields must be of the form $\mathbb{F}_{p^n}$ for some n , and these have $\mathbb{F}_p$ as a subfield. All fields of a given size are isomorphic. It is also convenient to define the *characteristic* of a field as the smallest number k for which $1 + 1 + \dots + 1$ (k times) equals 0. This never happens for infinite fields and they are said to have characteristic 0. Finite fields have characteristic p .

A. Polynomials over $\mathbb{F}$

Define $\mathbb{F}[X]$ as the set of all polynomials of the form

$$f(X) = f_0 + f_1X + \dots + f_mx^m \quad (1)$$

where $f_i \in \mathbb{F}$ and X is formally just a symbol. Here m is the degree of the polynomial and addition and multiplication of polynomials is ‘as usual’ with modular arithmetic for the coefficients. As multiplication tends to increase the degree of polynomials, all non-zero degree polynomials have no multiplicative inverse. Division is thus tricky and for this reason $\mathbb{F}[X]$ is an algebraic structure called a *ring* rather than a field. A polynomial is called *monic* if its highest coefficient $f_m = 1$.

A polynomial $g(X)$ is called a *divisor* of $f(X)$ if $f(X) = q(X)g(X)$ for some quotient $q(X)$. Trivial divisors are 1 and f itself. Slightly more specifically a divisor is a *factor* if it is monic and non-trivial. A $m > 0$ degree polynomial that has no factors is called *irreducible*. Remember the coefficients of the factors must be in $\mathbb{F}$, for finite fields no nasty square roots are allowed! There is a *unique factorisation* theorem that we shall not prove: every polynomial in $\mathbb{F}[X]$ may be written as a unique product of monic irreducible polynomials (times a constant). These monic irreducible polynomials, sometimes called prime polynomials, are analogous to the primes in $\mathbb{Z}$.

Consider a monic $g(X)$ of degree m . Any $f(X)$ can be decomposed as

$$f(X) = q(X)g(X) + r(X) \quad (2)$$

for unique q and remainder r , with $\deg(r) < m$. With this we can define mod- $g(X)$ addition and multiplication of polynomials. That is, the result of operations modulo $g(X)$ are in terms of the remainders. Note that for coefficients in $\mathbb{F}_p$ the size of the remainder set generated by mod- $g(X)$ operations is $|\mathbb{F}_p|^m$.

B. Fields $\mathbb{F}_{g(X)}$

With modulo $g(X)$ operations we can construct a field out of the set of remainder polynomials. Let $\mathbb{F} = \mathbb{F}_p$ and $g(X) \in \mathbb{F}_p[X]$ be a degree m irreducible, monic polynomial. The set of remainder polynomials under mod- $g(X)$ addition and multiplication has p^m elements of the form

$$r_0 + r_1X + \dots + r_{m-1}X^{m-1}, \quad r_j \in \mathbb{F}_p \quad (3)$$

which (one can prove) forms a field. As all finite fields are isomorphic this is *the* finite field $\mathbb{F}_{p^m}$.

For example, consider $m = 2$ polynomials over $\mathbb{F}_2$. The polynomial $g(X) = X^2 + X + 1 \in \mathbb{F}_2[X]$ is irreducible and monic and so we can form the remainder set. These are degree 0 and 1 polynomials, i.e. $\{0, 1, X, X + 1\}$ which form $\mathbb{F}_{2^2}$. This is closed as for example $X \cdot X$ modulo $(X^2 + X + 1)$ is $X + 1$.

C. Roots and Cyclicity of $\mathbb{F}_q^*$

If $f(x) \in \mathbb{F}[X]$ has a factor $X - a$ then a is a *root* of $f(X)$, i.e. $f(a) = 0$. The fundamental theorem of algebra states that a monic polynomial of degree k can have at most k roots in $\mathbb{F}$. If $\mathbb{F} = \mathbb{C}$ then it has exactly k roots.

We now take another view on the field $\mathbb{F}_q$, for some finite q . We cast aside our knowledge that $q = p^n$ and ignore polynomials for now. Define $\mathbb{F}_q^* = \mathbb{F}_q \setminus \{0\}$ and consider any element $\beta \in \mathbb{F}_q^*$. Keep multiplying it by itself $\beta, \beta^2, \dots$, eventually we must return to 1 (as this is a finite field). This thus generates a cyclic group $\{1, \beta, \beta^2, \dots, \beta^{l-1}\}$ whose size l must divide the abelian group $(\mathbb{F}_q^*, \cdot)$ by Lagrange’s theorem. Thus $\beta^l = 1$ and by extension

$$\beta^{q-1} = 1 \text{ for all } \beta \in \mathbb{F}_q^*. \quad (4)$$

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Thus all of the $\beta \in \mathbb{F}_q^*$ are distinct roots of the polynomial $X^{q-1} - 1 \in \mathbb{F}_q[X]$. Equivalently, including 0, all $\beta \in \mathbb{F}_q$ are roots of $X^q - X$ which can therefore be factored as

$$X^q - X = \prod_{\beta \in \mathbb{F}_q} (X - \beta). \quad (5)$$

Some β are special in that they generate the whole group (i.e. $l = q-1$). These are called primitive elements which we conventionally denote α . At least one always exists. Consequently a finite field can be written

$$\mathbb{F}_q = \{0, 1, \alpha, \alpha^2, \dots, \alpha^{q-2}\} \quad (6)$$

where α is primitive. $\mathbb{F}_q^*$ is a cyclic group of order $q-1$ generated by α . For example $\mathbb{F}_{2^4} = \{0, 1, \dots, \alpha^{14}\}$ with elements $\{\alpha^5, \alpha^{10}\}$ of order 3, $\{\alpha^3, \alpha^6, \alpha^9, \alpha^{12}\}$ of order 5. The remaining 8 elements are primitive (order 15).

D. $\mathbb{F}_q^*$ as Polynomials

We previously had the remainder field $\mathbb{F}_{g(X)}$ of polynomials with coefficients in $\mathbb{F}_p$. We have just constructed a general field $\mathbb{F}_q$ in terms of primitive elements. The link between the two viewpoints is the following theorem, stated without proof:

Every finite field $\mathbb{F}_q$ of characteristic p is isomorphic to the polynomial remainder field $\mathbb{F}_{g(X)}$ derived from an m^{th} degree monic irreducible polynomial $g(X) \in \mathbb{F}_p[X]$. By size, therefore, $q = p^m$ for some m .

As before the polynomial $X^{p^m} - X$ may be factorised into its roots $x - \beta$ for every $\beta \in \mathbb{F}_{p^m}$. Further however we can prove that every m^{th} degree (monic) irreducible polynomial in $\mathbb{F}_p[X]$ divides this polynomial. Thus we also have a guaranteed another factorisation of $X^{p^m} - X$.

The polynomial $X^{p^m} - X \in \mathbb{F}_{p^m}[X]$ can be written as a product of all monic irreducible polynomials in $\mathbb{F}_p[X]$ of degree m or a number that divides m , without repetition.

Irreducible polynomials are all very good, but can we do better? An irreducible polynomial $g(X) \in \mathbb{F}_p[X]$ of degree m that has a primitive element, $\alpha \in \mathbb{F}_{p^m}$, as a root is known a *primitive polynomial*. More formally primitive polynomials are minimal polynomials of primitive elements. Given a primitive α , its *minimal* polynomial is the lowest degree monic polynomial (in $\mathbb{F}_p[X]$) that has α (in $\mathbb{F}_{p^m}$) as a root. It is important to stress that primitive $g(X)$ is not a field element; it generates the remainder field $\mathbb{F}_{g(X)}$ of degree $m-1$ polynomials. Under the isomorphism $\mathbb{F}_{p^m} \cong \mathbb{F}_{g(X)}$ the primitive element $\alpha \longleftrightarrow X$ modulo $g(X)$. We care a lot about primitive polynomials as all of their roots are primitive elements and thus generate the group: indeed (unproven) if the

original α is a root, so is $\alpha^p, \alpha^{p^2}, \dots, \alpha^{p^{m-1}}$ also all primitive. For a given m there are $\varphi(p^m - 1)/m$ different primitive polynomials to choose from, where $\varphi(n)$ is Euler's totient function (the number of integers coprime to n). That is, there are $\varphi(p^m - 1)$ primitive elements, and each primitive polynomial corresponds to m of these (m roots). Examples of binary primitive polynomials are given in Table I below.

Degree (n)	$g(x)$
1	$x + 1$
2	$x^2 + x + 1$
3	$x^3 + x + 1$
4	$x^4 + x + 1$
5	$x^5 + x^2 + 1$
6	$x^6 + x + 1$
7	$x^7 + x + 1$
8	$x^8 + x^6 + x^5 + x + 1$

TABLE I. Primitive polynomials over $\mathbb{F}_2$ of degree n (non-exhaustive)

E. Addition of Field Elements

So far we have talked a lot about multiplication. This is easy in terms of primitive elements, $\alpha^i \cdot \alpha^j = \alpha^{i+j}$, but what about addition $\alpha^i + \alpha^j$? This is α^k for some k , but what k ? To save laborious computations one usually looks up the result in a 'logarithm' table. For a primitive element α , define the *Zech logarithm* as

$$Z_\alpha(n) = \log_\alpha(1 + \alpha^n) \quad (7)$$

or more properly the solution to $\alpha^{Z_\alpha(n)} = 1 + \alpha^n$. This enables addition to be done quickly

$$\alpha^i + \alpha^j = \alpha^i(1 + \alpha^{j-i}) = \alpha^{i+Z_\alpha(j-i)}. \quad (8)$$

For characteristic $p = 2$ (binary) fields this has an 'involution' type property

$$Z_\alpha(n) = m \iff Z_\alpha(m) = n \quad (9)$$

which is easily proved by noting

$$\alpha^{Z(Z(n))} = 1 + \alpha^{Z(n)} = 1 + 1 + \alpha^n = \alpha^n \pmod{2}. \quad (10)$$

F. Vector Spaces and Traces

The final element we must cover before turning to M-sequences is the concept of finite field traces. In linear algebra the trace is the (basis independent) sum of a matrix's diagonal elements, but how do we get to there

from finite fields? The answer really has been staring us in the face all along: $\mathbb{F}_{p^m}$ naturally forms a vector space with coefficients in the subfield $\mathbb{F}_p$. This is most easily seen from the polynomial remainder field interpretation, $\mathbb{F}_{g(X)} \cong \mathbb{F}_{p^m}$. The ‘vectors’ are the polynomials with normal polynomial addition and scalar multiplication (mod- p). Note that while $\mathbb{F}_{p^m}$ can be viewed as a vector space its structure is richer, being a field, due to the ability to multiply the ‘vectors’.

The set $\{1, X, X^2, \dots, X^{m-1}\}$ forms a basis for this space with specific elements represented as tuples / coordinates $(a_0, a_1, \dots, a_{m-1})$ with $a_i \in \mathbb{F}_p$. We can then define linear maps on this space, represented by matrices. Consider the map which is a simple multiplication by some element $\beta \in \mathbb{F}_{p^m}$, i.e. $x \rightarrow \beta x$. Denote this as $m_\beta : \mathbb{F}_{p^m} \rightarrow \mathbb{F}_{p^m}$. This takes an input tuple and maps it to another tuple and thus has a corresponding matrix M_β . For example consider $\mathbb{F}_{2^3}$ with $g(X) = X^3 + X + 1$ and multiplication by $\beta = 1 + X$. This maps the basis

$$\begin{aligned} 1 &\longrightarrow 1 + X \\ X &\longrightarrow X + X^2 \\ X^2 &\longrightarrow X^2 + X^3 = X^2 + X + 1, \end{aligned} \quad (11)$$

which has matrix representation

$$M_\beta = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}. \quad (12)$$

Note that in this case $g(X)$ was primitive but only irreducibility is required here. With this all in mind we can define the *trace of a finite field element* $\beta \in \mathbb{F}_{p^m}$ as

$$\text{Tr}_{\mathbb{F}_{p^m}/\mathbb{F}_p}(\beta) \equiv \text{trace}(M_\beta) \quad (13)$$

that is the trace of the linear map $x \rightarrow \beta x$ in the normal linear algebra sense. Correspondingly this is a linear operation. For the above example the trace is $3 = 1 \pmod{2}$. The subscript $\mathbb{F}_{p^m}/\mathbb{F}_p$ is there to remind us that the trace is a map from the full $\mathbb{F}_{p^m}$ down to the prime subfield $\mathbb{F}_p$. We shall omit this henceforth. While we shall not prove it one can show that (13) can alternatively be written as a sum of ‘conjugates’ $\sum_{i=0}^{m-1} \beta^{p^i}$.

II. M-SEQUENCES

We now know enough to properly define M-sequences. In the communications literature finite fields are usually referred to as *Galois Fields* and denoted $\text{GF}(p^n)$ rather than $\mathbb{F}_{p^n}$. The field $\text{GF}(p^n)$ is also often referred to as the *extension* of the base field $\text{GF}(p)$.

A. From Galois Fields

Take the binary field $\text{GF}(2^n)$ parameterised in terms of some primitive element α and its corresponding mini-

mal (primitive) polynomial $g(X)$. M-sequences are constructed in terms of the trace of the extension field back into $\text{GF}(2) = \{0, 1\}$. An M-sequence $s[i]$ is defined as

$$s[i] \equiv \text{Tr}(\alpha^i) \in \{0, 1\}, \quad i = 0, 1, \dots \quad (14)$$

which, as α is primitive, generates the entire field apart from the zero element. Thus M-sequences are of length $L = 2^n - 1$, periodically repeating after this as $\alpha^{2^n - 1} = 1$. It is important that α is primitive, other elements would generate a sequence but it would not be of maximum length!

In digital applications M-sequences are typically defined as either

$$m[i] \equiv (-1)^{s[i]} \quad \text{or} \quad m[i] \equiv 2s[i] - 1, \quad (15)$$

mapping to ± 1 rather than $\{0, 1\}$. The two definitions differ by a global minus sign. Cyclic shifts of M-sequences just correspond to multiplication by some field element $\beta = \alpha^\tau$, that is $s[i + \tau] = \text{Tr}(\beta \alpha^i)$. This gives the curious property that the product of an M-sequence with a shifted version of itself is *also* an M-sequence at a *different* offset. Using the first definition and the linearity of the trace

$$\begin{aligned} m[i] m[i + \tau] &= (-1)^{s[i] + s[i + \tau]} \\ &= (-1)^{\text{Tr}(\alpha^i) + \text{Tr}(\alpha^{i + \tau})} \\ &= (-1)^{\text{Tr}(\alpha^i (1 + \alpha^\tau))} \\ &= (-1)^{\text{Tr}(\alpha^{i + Z_\alpha(\tau)})} \\ &= m[i + Z_\alpha(\tau)], \end{aligned} \quad (16)$$

the offset is in terms of the Zech logarithms defined in Section IE.

B. As Recurrence Relations

Consider the n^{th} order primitive polynomial

$$g(X) = g_0 + g_1 X + \dots + g_{n-1} X^{n-1} + X^n, \quad (17)$$

arising from the primitive element α . We have that $g(\alpha) = 0$ and thus $0 = \sum_i g_i \alpha^i$. Premultiplying by some offset α^τ and taking the trace we get

$$0 = \sum_{i=0}^n g_i s[i + \tau] \pmod{2} \quad (18)$$

or rearranged, $s[n + \tau] = -\sum_{i=0}^{n-1} g_i s[i + \tau]$ for any τ . We see that $s[n]$ is determined entirely by the n previous values $s[0], \dots, s[n-1]$ and by extension

$$s_i = -(g_{n-1} s_{i-1} + g_{n-2} s_{i-2} + \dots + g_0 s_{i-n}) \pmod{2} \quad (19)$$

changing notation for brevity. M-sequences can therefore equivalently be defined as an order n *recurrence relation* in terms of the primitive polynomial coefficients.

Initialising with n binary values specifies the whole sequence. Barring $(0, \dots, 0)$ which will remain at 0 there are $2^n - 1$ possible initial conditions which generate cyclic shifts of the same M-sequence. Different M-sequences arise from different primitive polynomials, at order n there are $\varphi(2^n - 1)/n$ of these. Explicitly for $n = 2, 3, \dots$ there are 1, 2, 2, 6, 6, 18, 16, $\dots$ unique sequences. Note that the initial condition is often quoted as an integer seed arising from interpreting $(s_{n-1}, \dots, s_0)$ as a binary number. The same trick can also be done with the polynomial coefficients.

C. From Shift Registers

At any point i in the sequence we can define the current *state* as $(s_i, s_{i-1}, \dots, s_{i-n+1})$, i.e. all the values required to calculate s_{i+1} . Each step *shifts* the state right

$$(s_i, s_{i-1}, \dots, s_{i-n+1}) \longrightarrow (s_{i+1}, s_i, \dots, s_{i-n+2}) \quad (20)$$

where the new value s_{i+1} comes from ‘feedback’ from the previous values in a linear manner (19). This defines an (additive) linear-feedback shift register (LFSR) of length n . Over the course of the sequence all possible states of the shift register will be realised.

A graphical representation of a specific order 4 sequence with primitive polynomial $X^4 + X + 1$ is shown in Figure 1. Let $\{a_3, a_2, a_1, a_0\}$ denote the register values, initialised at $\{1, 1, 1, 1\}$. Now shift everything to the right, sending $a_0 = 1$ to the output on the right. This is the first term in our order 4 M-sequence. As we successively shift right the first 4 outputs are 1, 1, 1, 1, i.e. the initial state of the register. The register values therefore satisfy $s[i] = a_r[i - r]$. For this primitive polynomial only g_0 and g_1 are non-zero meaning $s_i = -s_{i-3} - s_{i-4}$ or equivalently $s_{i+4} = s_{i+1} + s_i \pmod{2}$. In terms of register values this becomes $a_3[i + 1] = a_1[i] + a_0[i] \pmod{2}$ meaning the vacant space on the left of the register is filled with the XOR (modulo 2 addition) of what was previously in a_1 and a_0 as shown in the diagram. The first fill is therefore $1 \oplus 1 = 0$ and the second state of the register is $\{0, 1, 1, 1\}$. We then repeat the process, sending 1 to the output and another 0 to the far left. This gives the pattern

$\{1, 1, 1, 1\}$	OUT : 1
$\{0, 1, 1, 1\}$	OUT : 1
$\{0, 0, 1, 1\}$	OUT : 1
$\{0, 0, 0, 1\}$	OUT : 1
$\{1, 0, 0, 0\}$	OUT : 0
$\{0, 1, 0, 0\}$	OUT : 0
$\vdots$	$\vdots$

This continues on until all possible $2^4 - 1$ configurations of the register have been exhausted ($\{0, 0, 0, 0\}$ cannot

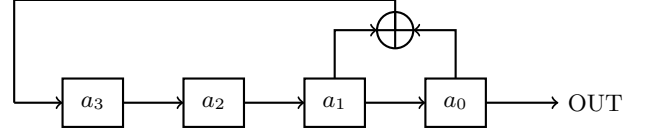


FIG. 1. An example linear feedback shift register of length 4. The $\oplus$ represents addition modulo 2.

be reached). The 15 output values form the order 4 m-sequence: 1, 1, 1, 1, 0, 0, 0, 1, 0, 0, 1, 1, 0, 1, 0. More complicated primitive polynomials will have multiple feedback ‘taps’ and the shift register set up is more involved.

D. Autocorrelation

The most famous property of M-sequences is their impulsive circular autocorrelation and thus flat power spectrum (by the Wiener-Khinchin Theorem). Consider again $\text{GF}(2^n)$ as a n -dimensional vector space over $\{0, 1\}$ with the M-sequence defined as a linear map (specifically the trace) on this space. M-sequence values of 0 therefore correspond to elements in the nullspace (kernel) of this map. The image of the map is $\{0, 1\}$, which has dimension 1 and so by the Rank-Nullity Theorem the kernel has dimension $n - 1$. Consequently a trace of 0 appears 2^{n-1} times and thus a trace of 1 also appears $2^n - 2^{n-1} = 2^{n-1}$ times: this is the *balance* property. However be careful to note that the field’s 0 element (the additive identity which has trace 0) is excluded from the sequence so the M-sequence itself has one more 1 than 0. Thus $\sum_{i=0}^{L-1} m[i] = -1$. This allows us to determine the autocorrelation of two length- L M-sequences. For $i \neq 0$

$$\begin{aligned} (m \otimes m)[i] &= \sum_{k=0}^{L-1} m[k]^* m[(k+i)_{\text{mod-}L}] \quad (21) \\ &= \sum_{k=0}^{L-1} (-1)^{\text{Tr}(\alpha^k)} \cdot (-1)^{\text{Tr}(\alpha^{k+i})} \\ &= \sum_{k=0}^{L-1} (-1)^{\text{Tr}(\alpha^k \alpha^{Z_\alpha(i)})} \\ &= \sum_{k=0}^{L-1} m[k + Z_\alpha(i)] = -1, \end{aligned}$$

as for any fixed i the $Z_\alpha(i)$ is just a cyclic shift. At $i = 0$ the sum reduces to $\sum_k m[k]^2 = \sum_k 1 = L$ and thus

$$(m \otimes m)[i] = \begin{cases} L, & i = 0, \\ -1, & \text{otherwise.} \end{cases} \quad (22)$$

Impulsive autocorrelations are an important property of truly random sequences. Of course M-sequences are deterministic, but for this reason they are referred to as a type of pseudorandom binary sequence (PRBS).